PROJECT OVERVIEW

Market Live

A live price dashboard for 80 instruments (IDX & US stocks, indices, crypto, gold including Antam bullion, commodities, and FX), with a trading plan whose rule was chosen through a 5-year backtest, for Indonesian retail investors.

Raihan Evanza · Fullstack & Mobile Developer · portfolio-raihan-evanza.vercel.app

ROLE

Personal project · design, frontend, serverless backend, data research, deployment

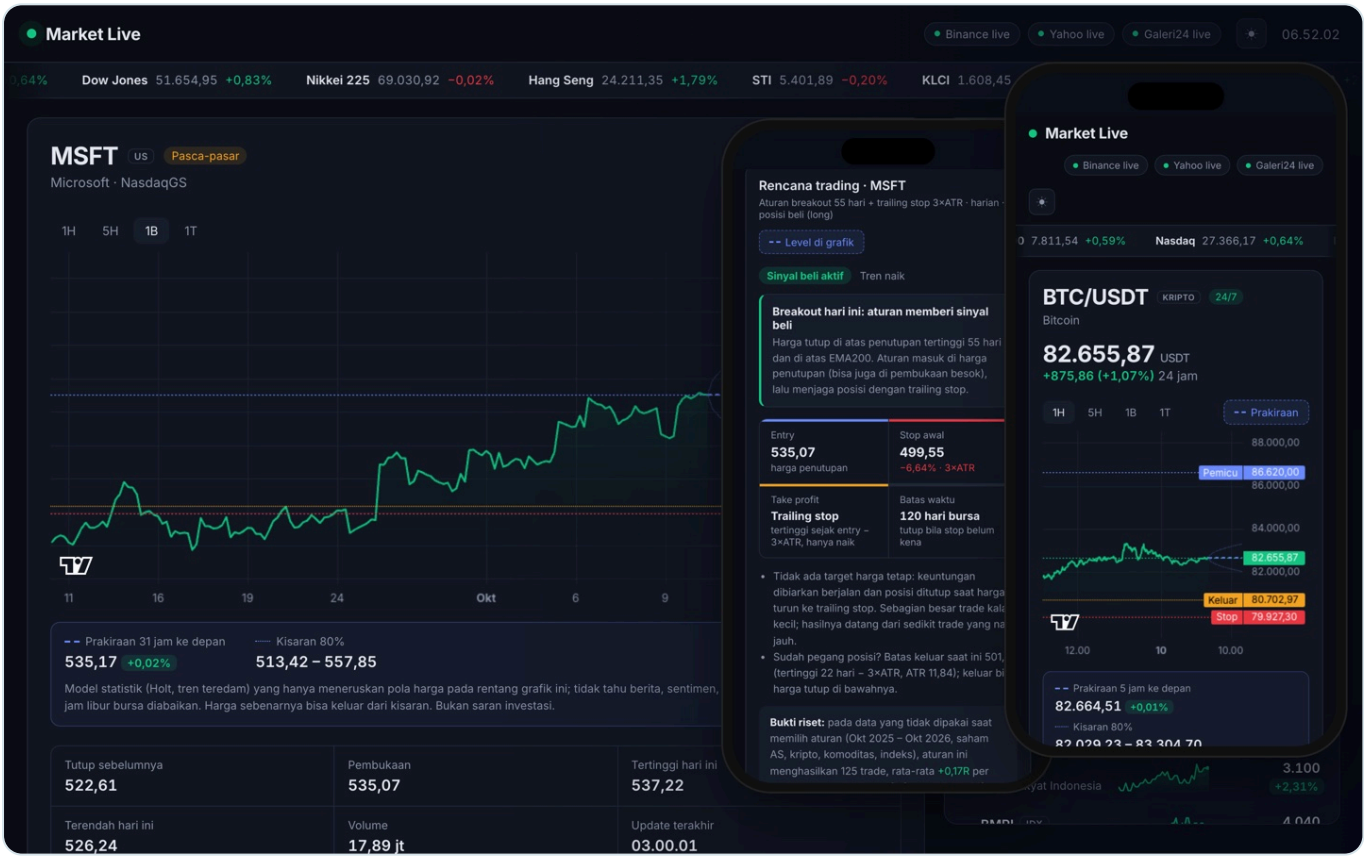
PLATFORMS

Web (desktop & mobile), no account needed

LINKS

Live demo: market-live-raihan-evanza.vercel.app

React 19 · Vite · Vercel Functions · Vercel Cron & Blob · TradingView Lightweight Charts · WebSocket · node:test



OVERVIEW

Retail investors in Indonesia usually juggle several apps: one for stocks, one for crypto, and another for Antam gold prices and the rupiah rate. Market Live brings 80 instruments (8 indices, 30 IDX stocks, 11 US stocks, 12 cryptocurrencies, 9 gold & commodity instruments, 10 FX pairs) into a single dashboard that updates live.

Every data source is free and needs no API key, and the app runs on the Vercel free tier. Crypto and spot gold stream in real time over a Binance WebSocket straight into the browser; stocks, FX, Antam prices, and news come through Vercel Functions cached at the CDN, and a daily job stores screener results in Vercel Blob.

Every instrument has a Trading plan panel built on a breakout rule chosen through a 5-year study with held-out test data. The research results are shown in the app as they are, including the fact that the rule is not yet statistically significant and is deliberately not applied to IDX stocks or FX.

KEY FEATURES

Real-time crypto & gold prices

12 cryptocurrencies (BTC, ETH, SOL, XRP, and more) and spot gold (PAX Gold) update on every tick over the Binance WebSocket, flashing green or red as prices change. Stocks, indices, commodities, and FX from Yahoo Finance refresh every 15 seconds.

An 80-instrument watchlist

Indices / Stocks / Crypto / Commodities / FX tabs, stocks split by exchange (IDX & US), search, 1-day sparklines, and favourites (★) saved in the browser. There is also a light/dark theme, and the selected symbol is kept in the URL (e.g. #BBCA.JK) so it can be shared.

Live charts & statistical projection

Charts from 1 day to 1 year whose last point moves live, plus the 52-week range. The Prakiraan (projection) button adds a trend line and an 80% band from a damped-trend Holt model, with a clear note that it is not a prediction.

Trading plan

A 55-day breakout rule with a 3×ATR trailing stop: today's status (active signal, waiting for a breakout, or below the EMA200), the trigger and initial stop drawn on the chart, and a position-size calculator from capital and % risk. IDX stocks, FX, and Antam gold get no signal because no rule passed testing in those markets.

5-year backtest per instrument

The same rule is simulated on the instrument being viewed: equity curve, win rate, average R, drawdown, recent trades, and a comparison with buy-and-hold. Example for NVDA: 19 trades at +0.74R per trade, while buy-and-hold over the same period gained far more.

Signal screener & forward test

The screener ranks every instrument the rule covers by how close it is to its trigger level, with the distance computed from live prices. The forward test records every new signal since 10 Oct 2026 with frozen entry and stop levels, so its results cannot be adjusted afterwards.

Full Antam gold prices

Antam sell and buyback prices for 0.5–100 g, the sell–buyback spread, the premium over spot gold, and a daily history recorded by the app because Galeri24 does not provide one. Example on 9 Oct 2026: 1 g sold at Rp2,611,000, an 8.5% premium over spot and an 11.3% spread to buyback.

Converter

Converts between rupiah, 8 foreign currencies, BTC/ETH, spot gold per gram, and Antam gold (sell or buyback price), all at live prices.

Market movers, comparison, dividends

Top 5 gainers and losers per market with an up/flat/down bar, a comparison chart of up to 6 instruments in % change (1 month to 5 years), and 12-month dividend yield plus a ranking of the highest-yielding IDX and US stocks.

Market news

News from the CNBC Indonesia, Kontan, and Antara RSS feeds, refreshed every 10 minutes, with a filter for stories related to the instrument being viewed.

ARCHITECTURE & DATA FLOW

Binance (WebSocket + REST)
Crypto & PAX Gold, accessed directly from the browser

→

Yahoo · Galeri24 · RSS
Stocks, indices, FX, dividends; Antam table; news

→

Vercel Functions + Cron
/api/yahoo, history, antam, news, dividends, data; daily job at 04:30 UTC

→

Vercel CDN · Blob
s-maxage cache 15 s–12 h; daily job results

→

Browser (React)
Merges quotes; charts, projection, trading plan & backtest

- Binance allows public CORS and WebSocket access, so crypto and spot gold flow from Binance straight to the browser with no server in between.
- Yahoo Finance sends no CORS headers, Galeri24 has no API, and the news RSS feeds need parsing, so all of them are fetched in Vercel Functions (Singapore region). The CDN caches each response according to how fast the

data changes: quotes 15 seconds, chart history 1 minute–1 hour, Antam 30 minutes, news 10 minutes, dividends 12 hours.

- Vercel Cron runs a daily job at 04:30 UTC (about 11:30 Jakarta time): it computes the screener and forward test from finalized daily bars, then records that day's Antam price. The results are stored as JSON in a private Vercel Blob store and read through /api/data.
- The trading plan and backtest are computed in the browser from 5 years of daily history, using the same simulator as the research scripts and the daily job.

STACK BY LAYER

Frontend	React 19 (useReducer, custom hooks) · Vite · TradingView Lightweight Charts 5 · Plain CSS (responsive, light/dark)
Backend	Vercel Functions (Node.js 20+) · Vercel Cron · Vercel Blob (private) · Vercel CDN cache (s-maxage, stale-while-revalidate)
Data sources	Binance public WebSocket & REST · Yahoo Finance (chart & spark endpoints) · Galeri24 / Pegadaian (HTML scraping) · RSS: CNBC Indonesia, Kontan, Antara
Quality & research	node:test unit tests · Node.js backtest scripts

TECHNICAL DECISIONS

1 Binance straight from the browser, Yahoo through serverless

Vercel Functions cannot act as a WebSocket server, but Binance lets browsers connect directly, so crypto stays real-time with no server at all. Yahoo blocks browser access, so it is polled through Functions; since Yahoo's IDX data is already about 10 minutes delayed, a 15-second poll is barely different from streaming.

2 CDN caching, and polling that pauses in hidden tabs

One /api/yahoo call fetches 65 symbols in 4 batch requests and is cached for 15 seconds (s-maxage=15), so no matter how many visitors there are, Yahoo is hit roughly once per 15 seconds per region. A tab left open uses about 170 thousand requests a month out of a free quota of 1 million, so polling stops on the visibilitychange event when the tab is hidden.

3 Antam from Galeri24, with a self-recorded history

logammulia.com blocks server requests through Cloudflare, so Antam prices come from Galeri24 (a Pegadaian subsidiary) by parsing the table in its HTML. Galeri24 keeps no history, so the daily job records the price to Blob every day; meanwhile the Antam chart is estimated from spot gold per gram times today's premium.

4 A deliberately simple Holt projection

A damped-trend Holt model runs on log prices in the browser; its parameters are picked from 96 combinations by the lowest one-step-ahead error, in under 1 millisecond. The 80% band widens with distance ($\sigma\sqrt{h}$), and the horizon is only 20% of the chart range, so the line is not read as a certainty.

5 Trading rule chosen with held-out test data

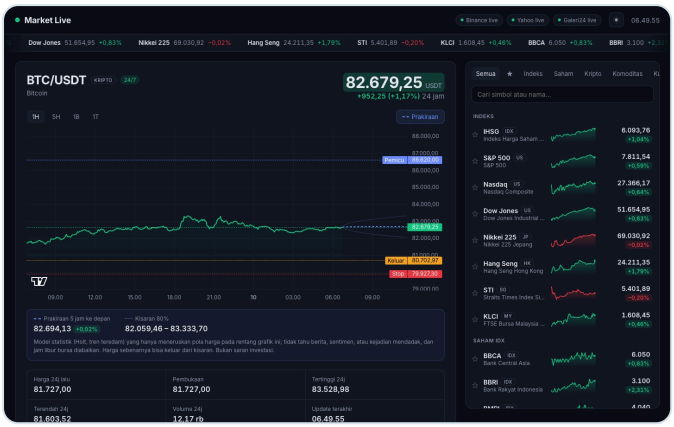
Five years of daily data (Oct 2021–Oct 2026, 77 instruments) were split into train, validation, and a final test opened only after the rule was locked; 11 candidates were tested with per-market trading costs. The old rule lost −0.17R per trade (5,456 trades), while the 55-day breakout with a 3×ATR trailing stop made +0.10R in train and +0.17R in validation. In the final test it made +0.04R across all markets and +0.17R (125 trades, profit factor 1.50, t = 1.59) on the markets that get signals.

6 One simulator for research, app, and forward test

The research scripts, the in-browser backtest panel, and the daily job all use the same simulator file, and a unit test checks that a signal on a given day never uses later data. That way the numbers in the app cannot silently drift from the research results.

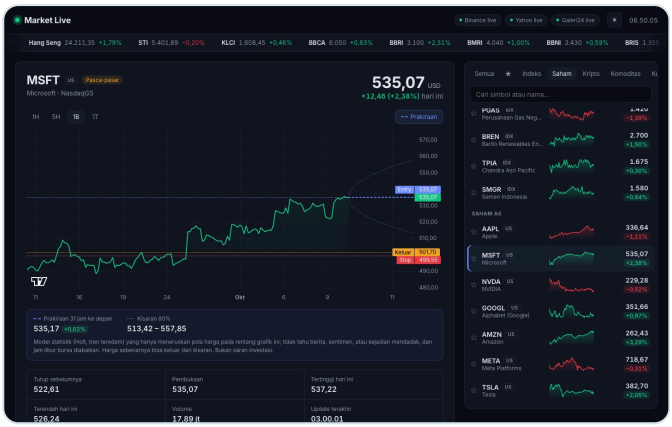
HOW TO USE IT

These steps follow the same order as the tutorial video.



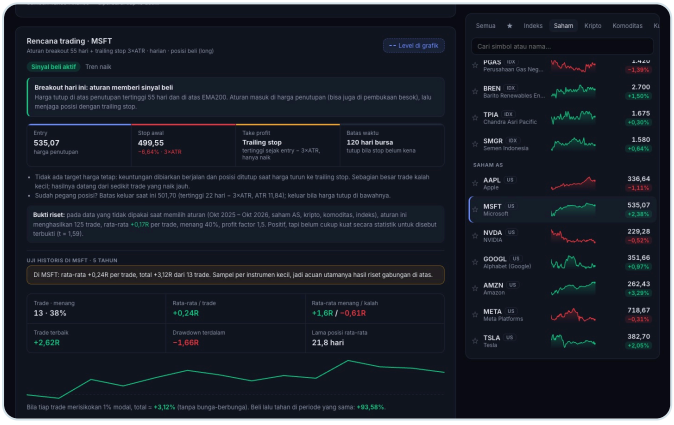
1 Open the dashboard

Open market-live-raihan-evanza.vercel.app. The page opens on BTC/USDT; watch the price flash green or red, the "Binance live", "Yahoo live", and "Galeri24 live" chips in the header, and the scrolling ticker tape.



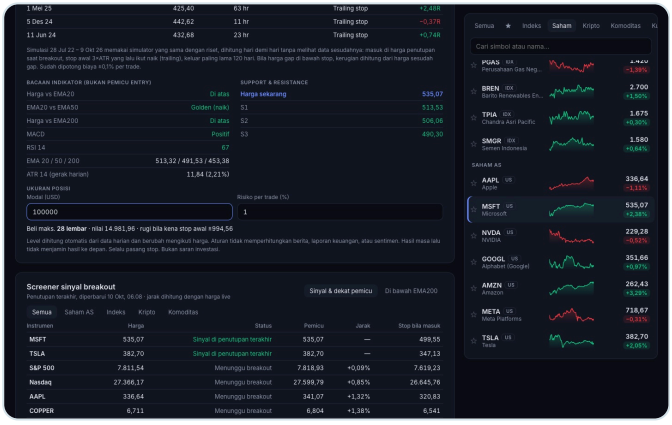
2 Pick a stock and read its chart

Click the "Saham" (Stocks) tab in the watchlist, then click MSFT (Microsoft). Click "1B" for one month of history; the dashed blue line at the end of the chart is the projection, with its 80% range shown below the chart.



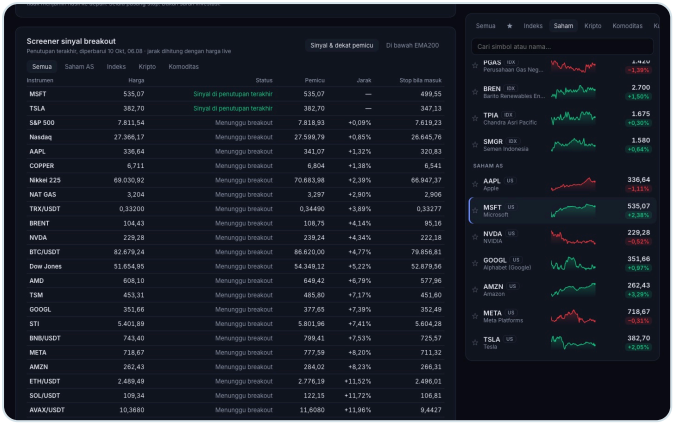
3 Read the trading plan

Scroll to the "Rencana trading" (Trading plan) panel. It shows today's status, the entry level, initial stop, and trailing-stop rule, plus a "Bukti riset" (research evidence) box with the final-test result and the note that it is not yet significant.



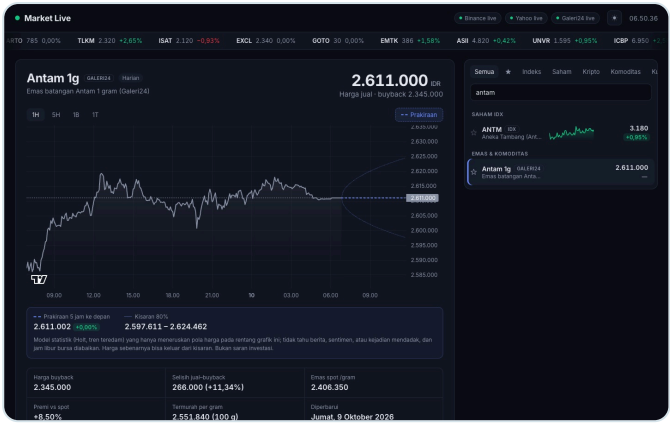
4 View the backtest and size a position

Below it, "Uji historis" (historical test) shows the 5-year equity curve, win rate, and a comparison with buy-and-hold. Enter 100000 in "Modal (USD)" (capital) to see the maximum number of shares at 1% risk.



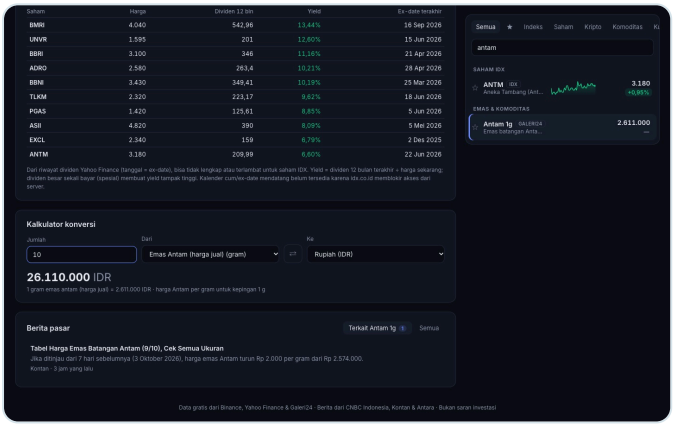
5 Check the screener and market movers

Scroll to "Screener sinyal breakout" (breakout screener) to see which instruments are closest to their trigger level. In "Penggerak pasar" (market movers), click "Saham AS" to see the top 5 US gainers and losers.



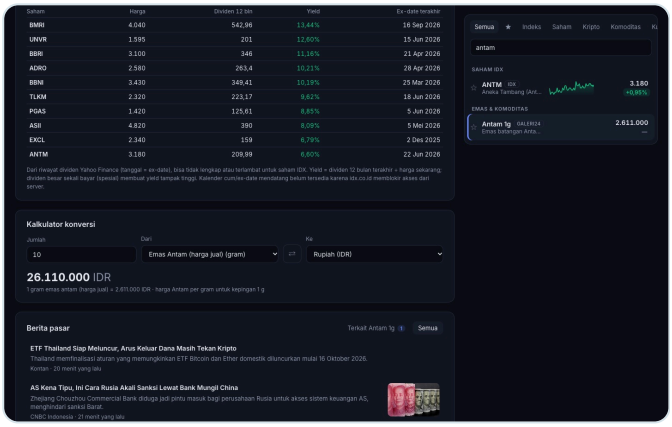
6 Check Antam gold prices

Click the "Semua" (All) tab, type "antam" into the watchlist search box, then click "Antam 1g". The panel shows sell and buyback prices for 0.5–100 g, the premium over spot gold, and the daily history; the trading plan explains why Antam gets no signal.



7 Convert gold to rupiah

Scroll to "Kalkulator konversi" (converter), enter 10 as the amount, then choose "Emas Antam (harga jual)" in the From field. The result appears in rupiah at today's price.



8 Read the news and switch theme

In "Berita pasar" (market news), click "Semua" (All) to see the latest stories from CNBC Indonesia, Kontan, and Antara. Finally, click the ☀/☾ button in the header to switch between light and dark themes.

LIMITATIONS & NOTES

- Not investment advice. All prices, projections, signals, and backtest results are for information and learning only, and should never be the sole basis for a decision to buy or sell.
- The breakout rule was positive in the final test (+0.17R per trade, 125 trades, 40% win rate), but with t = 1.59 the result is not statistically significant and could still be down to chance. The median trade is about −0.3R: profits come from a few trades that run far, and on stocks in a strong uptrend, buy-and-hold often does much better.
- On IDX stocks and FX, the rule was negative in both train and validation (e.g. IDX −0.04R and −0.11R), and no other candidate was consistently positive, so those markets deliberately get no signal. The results also carry survivorship bias, since the watchlist only holds large instruments that still exist today, and the simulation does not yet limit capital across many open positions.
- The forward test only started on 10 Oct 2026 and has no closed trades yet, so it cannot be used to judge the rule. Per-instrument backtests also contain only 10–20 trades, too few to draw conclusions from on their own.
- Yahoo Finance and Galeri24 have no official API: Yahoo's endpoints may change or throttle data-center IPs, and the Antam scraper can break if the site layout changes. On the Vercel free tier, the daily job can run up to an hour off schedule.

WHAT'S NEXT

- Judge the rule on 6–12 months of forward-test data, including RSI(2), which did better in the final test but lost in train. The rule is only switched if a candidate wins on truly new data.
- Portfolio-level simulation: a capital limit across many open positions and slippage beyond gaps, so backtest numbers come closer to real conditions.